

Late S-wave energy and decorrelation in Utah FORGE downhole microearthquakes: elastic tests of smooth, layered, fracture-zone and stochastic media

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Abstract

Downhole records of microearthquakes (MEQs) induced by the April 2022 stimulation at Utah FORGE contain long S-wave codas. 416 MEQs recorded at 4 kHz by two three-component sensors in well 56-32 are compared with 3-D finite-difference wavefields computed by reciprocity for smooth media (homogeneous, 50-m sonic-log layering, 3-D gradients, transverse isotropy), deterministic small-scale media (6-m sonic-log layering, 5-m-thick fracture zones) and lognormal stochastic media with power-law, exponential and Gaussian spectra. Moment tensors fitted to the direct waves ($P - 10$ ms to $S + 15$ ms) predict the 40–80 Hz energy from $S + 40$ to $S + 100$ ms. The smooth media under-predict this late energy by median factors of 44–82 at the upper and 12–15 at the lower sensor, robustly to windows, source constraints, source duration, event timing and a worst-case in-band tool resonance. Thin layers and fracture zones reduce it to factors of 11–19 (upper) and 2.4–3.4 (lower sensor); stochastic media span the observed late-energy distributions. In the early coda the waveforms of neighbouring events decorrelate with separation, as in the stochastic media, whereas the tested smooth and simple deterministic media stay coherent. In the pure late coda all media approach the noise level, and only the smooth media and the planar fracture zones remain more coherent than the data. The late energy implies an effective scattering strength

$\sigma_{\text{eff}} \approx 0.05\text{--}0.13$ (standard deviation of $\ln V$), depending on sensor and family; one domain-size test suggests that these are upper bounds, comparable to the sonic-log value of 0.045. At 20–80 Hz neither per-event nor common-grid statistical comparisons (event bootstrap, holdout) distinguish power-law from exponential or Gaussian spectra, and the sonic-log spectrum ($\beta = 0.85 \pm 0.07$ over 12–200 m) does not exclude a short correlation length. Within elastic modelling, the late S energy thus requires structure at the 5–10 m scale, and its decorrelation favours distributed 3-D heterogeneity without excluding fine layering, unless an unmodelled site response of comparable size is present; the spectral form is not resolved. Deterministic comparison against an unknown stochastic realisation favours smooth media, so medium-class selection should use wavefield statistics.

Keywords: EGS; Utah FORGE; microseismicity; seismic scattering; coda; random media; 3-D finite difference; downhole seismics

Introduction

Microseismic monitoring of engineered geothermal system (EGS) stimulation relies on a velocity model. In routine practice that model is homogeneous, layered or weakly anisotropic, smooth enough for ray theory to locate events and for a few source parameters to fit the direct waves. Everything after the direct S wave, the coda, is usually left unmodelled. That coda waves are generated by scattering on crustal heterogeneity is long established [1,2], and finite-difference simulations have shown how random media produce them [3]. Which heterogeneity produces the coda of a particular reservoir, and how strong it must be, remains much less clear, because near-source records with enough bandwidth to test it are rare.

Borehole data provide one candidate. Well-log fluctuation spectra in sedimentary, crystalline and volcanic rocks scale approximately as $S(k) \propto k^{-\beta}$ with $\beta \approx 1\text{--}1.5$ over centimetre-to-kilometre scales, and core porosity and permeability are related lognormally [4–6]. A medium with these statistics (a lognormal, spatially correlated "pink-noise" medium) is the basis of

geoflow imaging (GFI) [7,8], and Paper 1 of this series quantified such descriptors at Utah FORGE [9]. Whether the seismic wavefield of FORGE MEQs requires strong small-scale heterogeneity, and whether it can distinguish a power-law spectrum from spectra with a correlation length, has not been tested.

Here we use the 4-kHz records of two downhole sensors in well 56-32 during the April 2022 stimulation and 3-D elastic finite-difference (FD) wavefields for several hundred event–sensor paths per model. We ask three questions, in decreasing order of what the data can answer. (1) Do smooth media, or media with deterministic small-scale structure (thin layers, planar fracture zones), reproduce the late S-wave energy and its inter-event coherence? (2) If not, what effective scattering strength do stochastic media require? (3) Do the waveforms prefer a power-law (pink) spectrum over exponential or Gaussian spectra? We treat the first question as the main test, subject it to controls for window overlap, source, instrument ringing, event selection and numerical convergence, and report the second and third with their limitations. A methodological by-product is a synthetic demonstration that deterministic waveform comparison against an unknown stochastic realisation favours smooth models, so that medium-class selection must use wavefield statistics.

Data

Records and event selection

Table 1 Event selection

Step	Events	Magnitude; distance to sensor mid-point; share of stage 3
GES catalogue, April 2022 stimulation of 16A(78)-32	36 641	—
60-s SEG-2 files read (18–24 April 2022; 4 kHz, 6 channels)	3556 files	—
Largest events with $M \geq -1$ whose 1-s window lies inside a file (extraction cap, ordered by magnitude)	800	−0.44 [−0.61, 0.04]; 317 [262, 414] m; 83 %
Clean P onset on at least one sensor, hypocentre inside the FD interior	555	−0.44 [−0.60, −0.09]; 308 [261, 395] m; 92 %
Clean P onsets on both sensors: per-event coda test, D2, Φ	416	−0.44 [−0.60, −0.11]; 303 [261, 385] m; 97 %

150 highest P-wave SNR (subset of the original analysis; sensitivity only)	150	-0.40 [-0.60, -0.12]; 294 [267, 357] m; 96 %
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Median [10th, 90th percentile]. Stage 3 dominates at every step; stage 2 contributes 14 of the 416 events used, and stage 1 none.

Geo-Energie Suisse recorded the April 2022 stimulation of well 16A(78)-32 in 60-s SEG-2 files from monitoring well 56-32 and published a catalogue of 36 641 located events [10]. Fig. 1 shows the geometry, Fig. 2 example records, and Table 1 the selection steps. The only magnitude criterion is the extraction cap of 800 events; no step uses the coda. The per-event test uses all 416 events with clean P onsets on both sensors. The original analysis used the 150 events with the highest P-wave SNR. These have a slightly weaker, not stronger, observed coda than the remaining events (median $\log_{10}$ coda/S -1.33 against -1.12 at sensor A), so the selection did not inflate the coda.

Sensor calibration

The file headers carry no receiver coordinates. Automatic P onsets [11] on the two three-component sensors (channels 1–3, sensor A; 4–6, sensor B) were inverted with the catalogue hypocentres for sensor position (both sensors in one vertical well), clock offsets and V_P (L1 misfit). The solution places the sensors at depths of 2179 m (A) and 2522 m (B) with $V_P = 5808$ m/s. Event-bootstrap 95 % intervals are narrow (± 12 m for A, ± 81 m/s for V_P), but they understate the true uncertainty, because the catalogue locations were themselves computed with a smooth velocity model. Leaving out stimulation stage 3 moves both sensors 52 m deeper and lowers V_P by 8 %, which shows that depth, clock and velocity trade off structurally. We therefore regard sensor depths as uncertain by ~ 50 m and V_P by ~ 8 %, and test the effect of timing errors on the coda measurements directly (see below). The P travel-time residuals of neighbouring events are correlated up to ~ 40 m separation (Fig. 2c), which may reflect structure or correlated catalogue location error; we do not interpret them further.

Channels 1 and 4 carry only $\sim 3\%$ of the P energy and are not used; channels 2–3 and 5–6 behave as horizontal pairs, and all measurements use rotation-invariant horizontal energy.

Sonic log

The 56-32 sonic log (1950–2780 m, calliper-screened) gives a detrended $\ln V$ with $\sigma = 0.045$ at 6-m and 0.032 at 50-m averaging. Its spectral slope depends on band and estimator (Fig. 9). Over 12–200 m the multitaper estimate is $\beta = 0.85 \pm 0.07$ (taper jackknife), and across detrending orders, periodogram, Welch and multitaper estimators and three log segments it ranges from 0.57 to 1.21. Over 1–200 m it is 1.2–1.7. On log-binned spectra a power law is the best of four models over 12–200 m, but exponential and Gaussian autocorrelation spectra with correlation lengths of 2–4 m lie within $\Delta\text{AIC} = 3.4$ and 3.7. Over 1–200 m the Gaussian spectrum is excluded ($\Delta\text{AIC} = 26$), but the exponential one is not ($\Delta\text{AIC} = 2.2$). The log is thus compatible with power-law fluctuations, but a single 830-m vertical log cannot establish scale invariance or exclude a short correlation length. It is also a sample of intact rock along one line, not of the stimulated volume that the MEQ waves traverse.

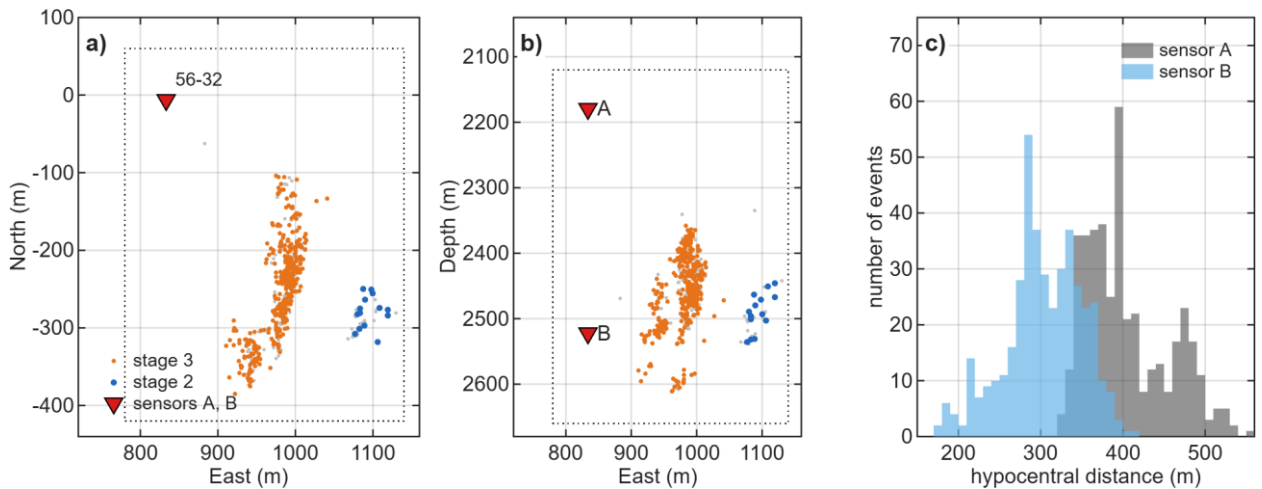


Fig. 1 Utah FORGE 2022 geometry: stimulation MEQs, 56-32 sensors and FD volume

a) Map view and b) east–depth section of the 416 analysed MEQs (orange: stage 3; blue: stage 2) and of the other catalogued events inside the FD interior (grey), with the calibrated sensors A and B in well 56-32 (red triangles) and the FD interior (dotted). Coordinates

relative to the 16A(78)-32 wellhead. c) Hypocentral distances of the 416 events to sensors A and B.

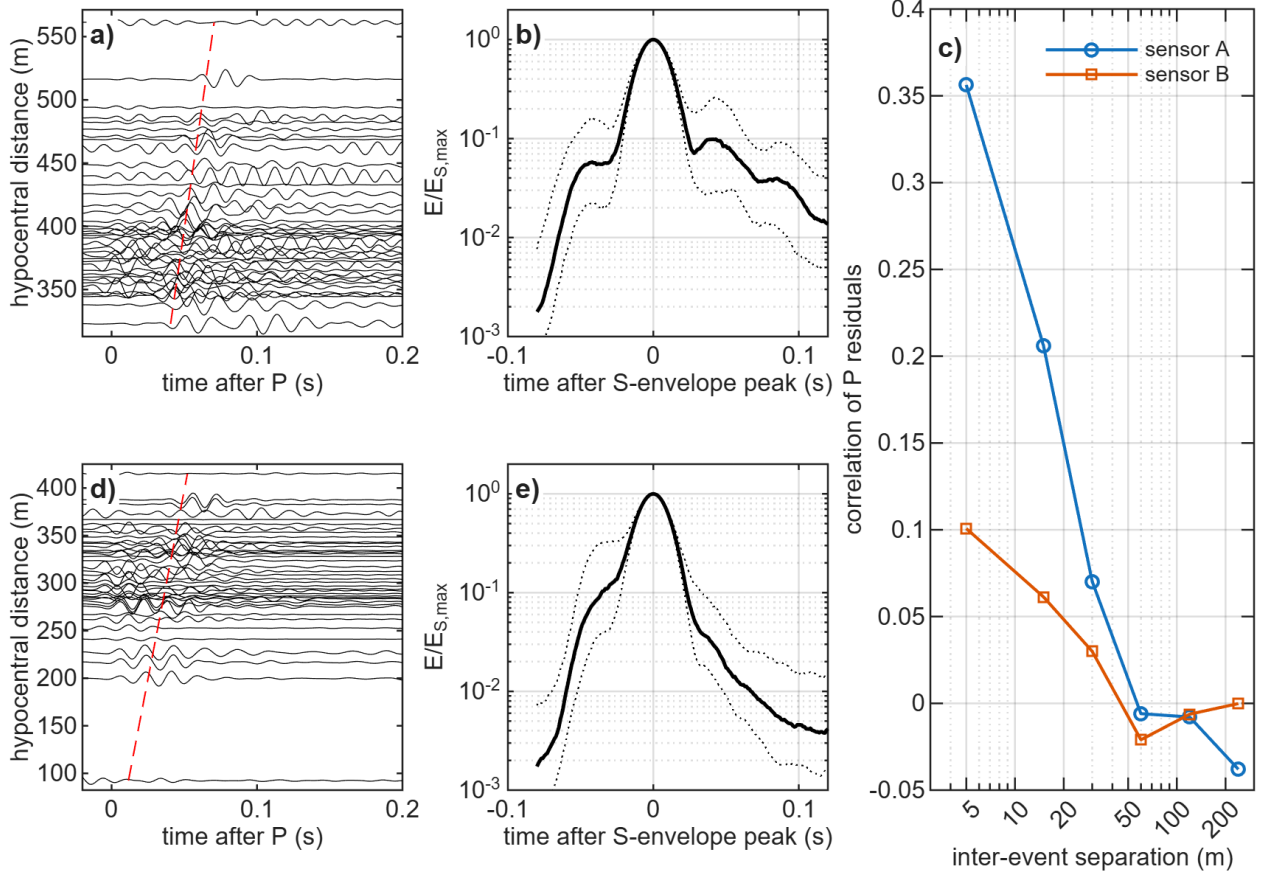


Fig. 2 FORGE 56-32 records and S envelopes

a, d) Horizontal records (one component, 40–80 Hz) of 40 events ordered by hypocentral distance for sensors A and B; the dashed line is the predicted S time. b, e) Median (solid) and interquartile range (dotted) of the horizontal S envelope, 40–80 Hz. c) Correlation of P travel-time residuals versus inter-event separation.

Methods

3-D elastic finite differences and reciprocity

We solve the velocity–stress equations on a staggered grid [12] with fourth-order spatial and second-order temporal differences [13] for a transversely isotropic medium with a vertical symmetry axis (VTI; stiffnesses from V_P , V_S , ρ and Thomsen ϵ , δ , γ [14]). The treatment of

125 anisotropy on the staggered grid follows Igel, Mora and Riollot [15], and the MATLAB
 126 implementation grew out of the acoustic codes of Igel [16], extended to 3-D elastic VTI
 127 media, moment-tensor and point-force sources and strain-rate recording. A Cerjan sponge
 128 [17] absorbs outgoing energy. The grid is $113 \times 137 \times 149$ nodes at 5 m (a $0.36 \times 0.48 \times 0.54$
 129 km interior plus 20-node sponges), with $\Delta t = 0.25$ ms, 1300 steps and a 45-Hz Ricker source.
 130 The analysis band is 20–80 Hz, with ≥ 5.4 grid points per S wavelength at 80 Hz even for the
 131 slowest truncated S velocity of the $\sigma = 0.18$ media (media with $\sigma = 0.22$ and 0.25 , which fall
 132 below 5 points, are shown only in Additional file 1). Three point-force runs per sensor record
 133 the strain-rate tensor at every event node, so that by reciprocity [18] the velocity at sensor
 134 component n from a moment tensor $\mathbf{M}$ at $\mathbf{x}_e$ is

$$v_n(\mathbf{x}_s, t) = M_{pq} \int_0^t \dot{\varepsilon}_{pq}^{(n)}(\mathbf{x}_e, \tau) d\tau. \quad (1)$$

135 Because only horizontal components are used, the two horizontal-force runs per sensor
 136 suffice. Against direct moment-tensor simulations in a heterogeneous VTI medium, the
 137 reciprocity synthetics agree with correlation 0.995 and amplitude ratio 0.995 (Additional file
 138 1: Fig. S1).

139 Coda amplitudes in a small domain could be affected by the grid, the domain edges and the
 140 sponge. We therefore simulated one band-limited stochastic medium ($\sigma = 0.13$, power-law
 141 spectrum with a 780-m outer scale, low-passed at 0.06–0.08 cycles/m so that it is identical on
 142 every grid) four times with sensor B, the same 358 events and the same double-couple
 143 mechanisms: on the reference 5-m grid, on a 2.5-m grid, in a domain extended by 150 m on
 144 every side, and with a sponge of 40 instead of 20 nodes. The coda measures of the three
 145 variants are compared event by event with the reference.

Media

All media have constant density $\rho = 2650 \text{ kg/m}^3$ and background $V_P = 5808 \text{ m/s}$, $V_P/V_S = 1.73$, except where stated. Four smooth media contain no structure below $\sim 50 \text{ m}$: a homogeneous medium (H), a layered medium from the 56-32 sonic log averaged in 50-m blocks (L), a medium with smooth 3-D gradients of $\pm 3 \%$ vertically and $\pm 2 \%$ horizontally across the box (GRAD3), and a homogeneous VTI medium with Thomsen parameters inverted from the P travel times (A). Two deterministic media contain small-scale structure: the sonic log sampled at its 6-m resolution on the 5-m grid (L6), and 12 planar fracture zones, one cell (5 m) thick, about 240 m tall, striking N25°E and dipping 75°, with V_P reduced by 10 % and V_S by 20 % (FZ, which also varies V_P/V_S).

The stochastic media are lognormal, $V(\mathbf{x}) = \bar{V} \exp[\sigma f(\mathbf{x})]$, where f is a zero-mean, unit-variance Gaussian random field obtained by filtering white noise $\hat{w}$ in the wavenumber domain. For the power-law (pink) field the filter is

$$\hat{f}(\mathbf{n}) \propto \frac{\hat{w}(\mathbf{n})}{(1 + |\mathbf{n}|)^p}, |\mathbf{n}| = L|\mathbf{k}|/2\pi, \quad (2)$$

where $\mathbf{n}$ is the dimensionless FFT index, L the box length along each axis and $\mathbf{k}$ the wavenumber. The roll-off wavenumber is therefore $k_0 = 2\pi/L$ (outer scale equal to the box, 0.6–0.75 km), and the power spectrum decays as $|\mathbf{k}|^{-2p}$ above it; $p = 1.5$ gives one-dimensional spectra $\propto k^{-1.05}$ and $p = 1.8$ gives $\propto k^{-1.48}$ along lines through the field. Exponential and Gaussian fields use the filters $(1 + k^2 a^2)^{-1}$ and $\exp(-k^2 a^2/8)$ with the correlation length a in metres. All fields are periodic over the FD box, which is larger than any source–receiver path. After scaling, $\ln V$ is truncated at $\pm 3\sigma$ ($\pm 2.5\sigma$ for $\sigma > 0.15$) without renormalisation. Truncation affects 0.25–1.2 % of cells and reduces σ by $\leq 1.1 \%$ (Additional file 1: Table S9). The truncated velocity range is wide: at $\sigma = 0.13$, V_P spans 3.9–8.6 km/s, and at $\sigma = 0.18$ 3.7–9.1 km/s. The extremes occupy a small fraction of the volume, but they show that $\sigma \geq 0.13$ is not a small perturbation.

In the default media V_P and V_S share one field, so V_P/V_S is constant. Two variants relax this at $\sigma = 0.13$: independent fields for V_P and V_S (VSD), and a V_S -dominated medium with $\sigma(\ln V_S) = 0.13$ and $\sigma(\ln V_P) = 0.065$ (VSO). Because σ absorbs any unmodelled contribution to the late energy, including density contrasts, anisotropic fracture scattering, intrinsic attenuation and site effects, we call the fitted value an *effective scattering-strength parameter* σ_{eff} within this isotropic model class, not a measured property of the rock.

Media were simulated in two sets (Table 2). The original set explores σ and p for the pink family. The revision adds a common grid for all three families: $\sigma = 0.09, 0.13$ and 0.18 ; $p = 1.5$ and 1.8 (pink), $a = 15$ and 50 m (exponential, Gaussian); two white-noise seeds each (11 and 33). Every family thus has the same number of candidates on the common grid. Fields of different families generated from the same seed share their phases and are strongly correlated (the pink and $a = 15$ m exponential fields with seed 33 by 96 %); this is a controlled comparison of spectral shape, not a test with independent fields.

Table 2 Media simulated

Model	Class	Parameters
H	smooth: homogeneous	V_P 5808 m/s, V_P/V_S 1.73
L	smooth: 1-D layered	56-32 sonic log, 50-m blocks
GRAD3	smooth: 3-D gradients	± 3 % vertical, ± 2 % E and N
A	smooth: homogeneous VTI	travel-time inverted ϵ, δ
L6	deterministic: 1-D layered	56-32 sonic log at 6-m resolution
FZ	deterministic: fracture zones	12 planar zones, 5 m thick, N25°E/75°, V_P -10 %, V_S -20 %
P4, P1, P3, P6, P9	pink, $p = 1.5$ (seed 11)	$\sigma = 0.02, 0.045, 0.09, 0.13, 0.18$
T1, R1, R2; R3; R6; R5	pink, $p = 1.5$	further seeds at $\sigma = 0.045; 0.09; 0.13; 0.18$
P5, P2; P7, P8; P11	pink, other p	$p = 1.2, 1.8$ at $\sigma = 0.045, 0.09$; $p = 1.8$ at 0.18
P12, P10	pink, $p = 1.5$	$\sigma = 0.22, 0.25$ (Additional file 1 only)
Common grid	pink, exponential, Gaussian	$\sigma = 0.09, 0.13, 0.18 \times (p = 1.5, 1.8 \text{ or } a = 15, 50 \text{ m}) \times \text{seeds } 11, 33$
VSD, VSO	pink, variable V_P/V_S	$\sigma = 0.13$; independent V_S field; $\sigma_{VS} = 0.13, \sigma_{VP} = 0.065$
R6c, R6d, E15c/d, G50c/d	synthetic truths	$\sigma = 0.13$, seeds 44 and 55 (shape-resolution test)
T2	synthetic truth	50-m layering + VTI ($\epsilon = 0.06, \delta = 0.03, \gamma = 0.06$)

All media: $\rho = 2650 \text{ kg/m}^3$. Stochastic media: lognormal, truncated at $\pm 3\sigma$ ($\pm 2.5\sigma$ for $\sigma > 0.15$). The common grid comprises 36 media (12 per family), 14 of which already existed in the original set.

Source and preprocessing

FD synthetics carry a 45-Hz Ricker source. To give data and synthetics the same source spectrum, each observed trace is convolved with the same Ricker wavelet. This is valid if the true source spectrum is flat in the analysis band, i.e. if corner frequencies exceed ~ 80 Hz. The S-window displacement spectra (raw data, 5–300 Hz) support this for most but not all events. The apparent corner, the first frequency above 50 Hz at which the spectrum falls 6 dB below its 30–50 Hz level, has a median of 128 Hz at sensor A and 90 Hz at B. Because it includes path attenuation, it is a lower bound on the source corner; 21 % (A) and 34 % (B) of events have apparent corners below 80 Hz. To bound the effect of finite source duration, we convolved the homogeneous and $\sigma = 0.13$ pink synthetics with Brune source-time functions with corners of 150, 100 and 60 Hz. Even the 60-Hz source raises the homogeneous coda-to-S ratio by only 0.35 log units (A) and changes the pink one by +0.03. Source duration therefore cannot close a deficit of 1–2 log units.

Per-event coda prediction test

For each event and medium, the moment tensor is fitted by linear least squares to the four horizontal traces (two sensors, 40–80 Hz) in the window $P - 10$ ms to $S + 15$ ms, with a per-sensor time shift of up to ± 6 ms and the tool azimuth and handedness of each sensor fixed from a previous grid search. The fitted moment tensor then predicts the whole record. After a guard band of 25 ms, the test quantity is the ratio of mean horizontal energy in $S + 40$ to $S + 100$ ms to the peak S energy ($S - 8$ to $S + 15$ ms), compared per event between prediction and observation as $\log_{10}(\text{predicted/observed})$. The fit and coda windows do not overlap. In the original version of this analysis the fit window extended to $S + 60$ ms and overlapped 43 % of the coda window; that version is superseded. The sensitivity analysis varies the end of the fit window ($S + 10$, 15, 20 and, for comparison, the original 60 ms) and the coda window ($S + 30$ –100, 40–100 and 50–110 ms). It also compares a full moment tensor, a deviatoric one and

211 a double couple (grid search over strike, dip and rake, with a least-squares scalar moment).
212 Condition numbers of the column-normalised design matrix and the ISO, CLVD and DC
213 fractions of the full solutions are reported. A timing check compares the observed S-envelope
214 peak with the predicted S time. The test is repeated on the events whose S peak lies within
215 ± 10 ms (and ± 5 ms) of the prediction on both sensors.

216 **Tests for instrument and borehole ringing**

217 A resonant sensor or borehole response $h(t)$ moves energy from the S peak into later
218 windows, so a coda-to-S ratio does not cancel it. We applied three tests. First, the S-window
219 spectra and the coda/S spectral ratio were examined for resonance peaks. In the analysis band
220 (30–120 Hz), any peak in the smoothed S spectrum is at most 0.35 dB above its surroundings.
221 The same measure detects the resonances near 330, 480 and 920 Hz (prominence up to 10
222 dB), which serve as a positive control. Second, in a null-ringing test the homogeneous
223 synthetics were convolved with the strongest single resonance (40–80 Hz, $Q = 5$ –40) whose
224 spectral peak does not exceed the observed in-band prominence, and the coda-to-S ratio was
225 recomputed. Third, the two sensors, which have different coupling and different in-band
226 peaks, are analysed separately throughout. No calibration shots, perforation shots or third
227 sensor were available for 2022, so an empirical transfer function could not be measured.

228 **Statistical comparison of media**

229 With two sensors the realisation of a random medium cannot be recovered, so stochastic
230 media are compared with the data through wavefield statistics. On the horizontal energy
231 envelope we measure three S-normalised features per event: the coda level $\log_{10}(E_{\text{coda}}/E_S)$ 30–
232 100 ms after the S peak, the coda decay rate and the rms S-pulse width. We use three sensor–
233 band combinations: A 20–40 Hz, A 40–80 Hz and B 40–80 Hz. B 20–40 Hz is excluded
234 because it carries a persistent non-event plateau. The misfit is

$$\begin{aligned}
\Phi = & \frac{1}{9} \sum_{c=1}^3 \sum_{j=1}^3 D_{\text{KS}}(F_{cj}^{\text{obs}}, F_{cj}^{\text{mod}}) \\
& + \frac{1}{3} \sum_{c=1}^3 \left[\frac{1}{N_\tau} \sum_{\tau=0}^{100\text{ms}} \left(\log_{10} \tilde{E}_c^{\text{obs}}(\tau) \right. \right. \\
& \left. \left. - \log_{10} \tilde{E}_c^{\text{mod}}(\tau) \right)^2 \right]^{1/2}
\end{aligned} \tag{3}$$

235 where D_{KS} is the two-sample Kolmogorov–Smirnov distance between the observed and
 236 synthetic distributions of feature j for combination c , and $\tilde{E}_c$ is the median post-peak S
 237 envelope. All nine KS terms and all three envelope terms are weighted equally, events are
 238 weighted equally, and events lacking a measurement in one combination are omitted from that
 239 combination only. The features are correlated, so Φ is a ranking statistic, not a likelihood. Its
 240 uncertainty is estimated by an event bootstrap (200 resamples, the same resample for every
 241 medium). Family comparisons use the best member of each family on the common grid per
 242 bootstrap sample, and a holdout test in which each family's best (σ , shape, seed) is chosen on
 243 half of the events and scored on the other half (20 random splits plus a near/far split by
 244 distance).

245 **Inter-event coda coherence**

246 For pairs of events recorded on the same sensor, D2 is the maximum normalised two-
 247 component cross-correlation (lag ± 8 ms) of a 40–80 Hz coda window, summarised as the
 248 median in bins of inter-event separation (0–5 to 80–160 m) and as the median over all pairs
 249 within 160 m. Three windows are used: S + 15 to S + 100 ms (early coda, including the tail of
 250 the S pulse), S + 40 to S + 100 ms (the window of the late-energy test) and S + 50 to S + 110
 251 ms. The coherence of noise alone is measured on the same records in a window S + 550 to S
 252 + 610 ms. Every synthetic record contains that event's own recorded noise at its observed

SNR, so the noise contribution is common to data and media. Pairs that share an event are not independent, so 95 % intervals come from a cluster bootstrap over events (each resampled event carries all its pairs). To test sensitivity to location error, the hypocentres used for binning were perturbed by isotropic Gaussian errors of 5, 10, 20, 30 and 50 m, for data and media alike. The all-pairs median does not depend on the binning and is therefore insensitive to location error.

Synthetic truth tests

The ability of the statistical comparison to select a medium class was tested with synthetic truths. Two truths were used in the main test: a pink medium (T1, $\sigma = 0.045$) and 50-m layering with VTI (T2). A further nine truths at equal $\sigma = 0.13$ (pink, exponential $a = 15$ m and Gaussian $a = 50$ m, seeds 33, 44 and 55) tested resolution of the spectral family. Synthetic records use double-couple mechanisms consistent with the FORGE stress field, and real 56-32 noise at each event's observed SNR.

Results

Smooth media under-predict the late S energy

Fig. 3 compares observed and predicted sensor-A seismograms for three events. The moment tensor fitted to the direct waves reproduces the P and S pulses in all media. In the homogeneous medium the prediction then decays to near zero within the guard band, whereas the observed traces keep oscillating through the coda window. Table 3 and Fig. 4 give the statistics over 416 events. The four smooth media under-predict the coda-to-S ratio by a median factor of 44–82 at sensor A and 12–15 at sensor B, and only 1–5 % (A) and 5–11 % (B) of events fall within a factor of three. Smooth 3-D gradients ($-1.67/-1.11$, A/B) and VTI anisotropy ($-1.91/-1.19$) alter the direct waves but add no late energy.

276 The two simple deterministic small-scale media tested change this substantially. The sonic log
 277 at its 6-m resolution (L6) gives $-1.28/-0.53$, and the thin fracture zones (FZ) give
 278 $-1.04/-0.39$: under-prediction by factors of 19 and 11 at sensor A but only 3.4 and 2.4 at
 279 sensor B, where 41 % and 45 % of events fall within a factor of three. Averaging the log in
 280 50-m blocks, as in the original analysis, therefore removed most of the late energy that the
 281 logged layering can generate. For the level of late energy, structure at the 5–10 m scale is the
 282 essential ingredient, whether it is layered, planar or random. Stochastic media span the
 283 observations: pink media with $\sigma = 0.02$ under-predict ($-1.20/-0.49$), $\sigma = 0.09$ is close at
 284 sensor A (-0.14) and $\sigma = 0.045$ at sensor B ($+0.02$), and the seed-11 stochastic media with $\sigma =$
 285 0.13 over-predict ($+0.28$ to $+1.12$). The best stochastic media bring 53 % (A) and 54 % (B) of
 286 events within a factor of three. The residual scatter reflects the unknown realisation. No single
 287 medium matches the median at both sensors; for the pink family, sensor B requires about half
 288 the σ of sensor A.

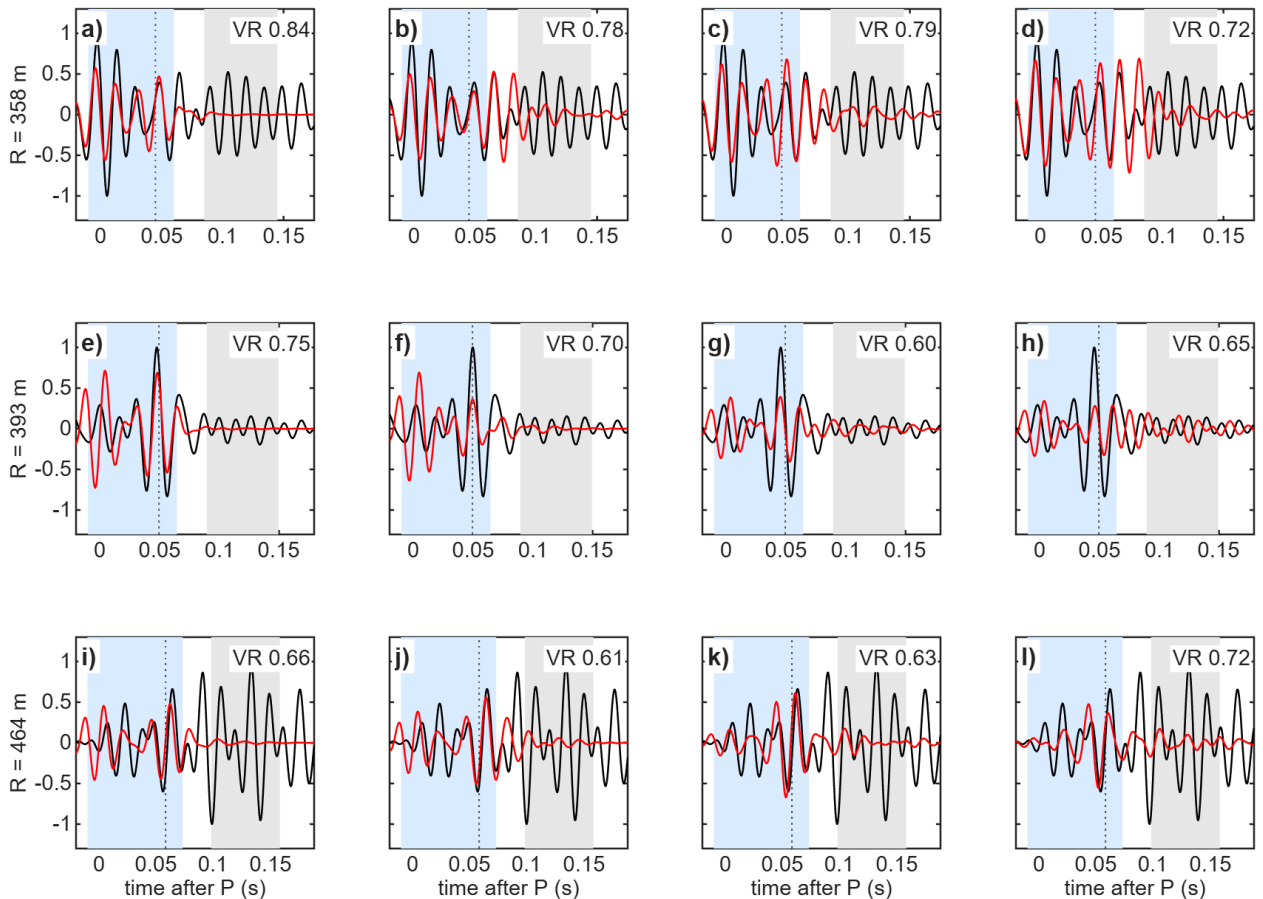


Fig. 3 Observed and predicted sensor-A seismograms with non-overlapping fit and coda windows

Horizontal component 1, 40–80 Hz, for three events at increasing distance (rows; distance on the left). Columns: homogeneous medium, deterministic fracture zones (FZ), pink media with $\sigma = 0.045$ and 0.09 . Black: observed; red: predicted with the moment tensor fitted in the light-blue window (P – 10 ms to S + 15 ms); grey: coda window (S + 40 to S + 100 ms); dotted: predicted S time. Variance reduction of the fit in each panel.

Table 3 Predicted versus observed late S energy for 416 FORGE events

Medium	Median $\log_{10}(\text{pred}/\text{obs}), A$	Median, B	% within $\times 3, A$	% within $\times 3, B$	VR
smooth: homogeneous	–1.80	–1.09	2	9	0.75
smooth: layered, 50 m	–1.64	–1.17	5	5	0.73
smooth: 3-D gradients	–1.67	–1.11	3	9	0.74
smooth: VTI	–1.91	–1.19	1	11	0.64
deterministic: layered, 6 m	–1.28	–0.53	15	41	0.73
deterministic: fracture zones	–1.04	–0.39	24	45	0.73
pink $\sigma = 0.02$	–1.20	–0.49	16	42	0.73
pink $\sigma = 0.045$	–0.74	+0.02	34	51	0.72
pink $\sigma = 0.09$	–0.14	+0.53	50	35	0.67
pink $\sigma = 0.13$	+0.31	+0.78	50	26	0.62
pink $\sigma = 0.18$	+0.61	+1.03	37	18	0.60
pink $\sigma = 0.13$, independent V_S	–0.13	+0.54	53	35	0.66
pink, V_S -dominated	+0.32	+0.79	49	24	0.62
exponential $a = 15$ m, $\sigma = 0.09$	–0.07	+0.61	51	33	0.68
exponential $a = 15$ m, $\sigma = 0.13$	+0.28	+0.94	50	22	0.64
exponential $a = 50$ m, $\sigma = 0.09$	–0.16	+0.39	48	41	0.65
exponential $a = 50$ m, $\sigma = 0.13$	+0.31	+0.71	45	27	0.59
Gaussian $a = 15$ m, $\sigma = 0.09$	+0.14	+0.79	48	22	0.69
Gaussian $a = 15$ m, $\sigma = 0.13$	+0.42	+1.12	43	15	0.64
Gaussian $a = 50$ m, $\sigma = 0.09$	–0.45	–0.35	39	44	0.62
Gaussian $a = 50$ m, $\sigma = 0.13$	+0.29	+0.54	45	33	0.56

Late energy = mean horizontal energy S + 40 to S + 100 ms divided by the S-peak energy, 40–80 Hz. Moment tensors fitted in P – 10 ms to S + 15 ms (no overlap with the coda window). VR = median variance reduction of the direct-wave fit. Stochastic media: seed 11; other seeds in Additional file 1.

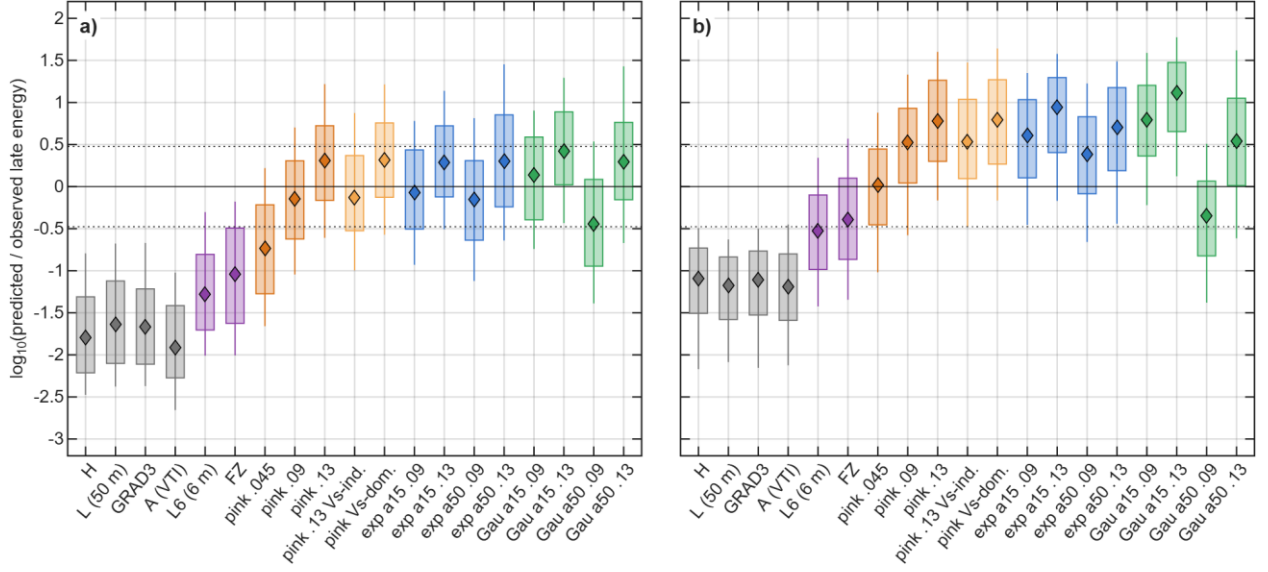


Fig. 4 Per-event predicted/observed late S energy for all media

a) Sensor A. b) Sensor B. 416 events, 40–80 Hz, windows as in Table 3. Boxes: interquartile range; whiskers: 10–90 %; diamonds: median; dotted lines: factor of three. Grey: smooth media; purple: deterministic small-scale media (6-m layering, fracture zones); orange: pink media (light orange: variable V_p/V_s); blue: exponential; green: Gaussian.

Robustness of the deficit

Table 4 Robustness of the late-energy deficit of smooth and deterministic media

Variant	Smooth, A	Smooth, B	L6 and FZ, A	L6 and FZ, B
Primary (fit to S + 15 ms; coda S + 40–100 ms)	–1.91 to –1.64	–1.19 to –1.09	–1.28 to –1.04	–0.53 to –0.39
Fit to S + 10 ms	–1.91 to –1.64	–1.20 to –1.07	–1.29 to –1.04	–0.52 to –0.33
Fit to S + 20 ms	–1.91 to –1.64	–1.19 to –1.08	–1.28 to –1.07	–0.53 to –0.39
Coda S + 50–110 ms	–2.21 to –1.99	–1.44 to –1.39	–1.46 to –1.33	–0.65 to –0.64
Coda S + 30–100 ms	–1.58 to –1.45	–1.05 to –0.99	–1.11 to –0.82	–0.48 to –0.19
Original overlapping fit (S + 60 ms), coda S + 30–100 ms	–1.59 to –1.45	–1.03 to –0.99	–1.12 to –0.93	–0.50 to –0.29
Deviatoric moment tensor	–2.04 to –1.69	–1.21 to –1.09	–1.33 to –1.19	–0.51 to –0.49
Double couple	–2.10 to –1.69	–1.25 to –1.11	–1.34 to –1.24	–0.58 to –0.51
150 highest-SNR events	–1.86 to –1.53	–1.26 to –1.09	–1.22 to –0.77	–0.65 to –0.33
S peak within ± 10 ms on both sensors (250 events)	–1.70 to –1.42	–1.17 to –0.98	–1.10 to –0.88	–0.53 to –0.35

S peak within ± 5 ms on both sensors (168 events)	-1.62 to -1.35	-1.20 to -0.98	-1.04 to -0.87	-0.55 to -0.42
Homogeneous + worst in-band resonance (null-ringing test)	-1.99 to -1.61	-1.33 to -0.96		
Homogeneous, Brune source $f_c = 60$ Hz (change)	+0.35	+0.35		

309 Median $\log_{10}(\text{predicted/observed late energy})$; range over the smooth media (H, L, GRAD3, A) and over the
310 deterministic small-scale media (L6, FZ). Null-ringing test: range over resonance frequencies 40–80 Hz and $Q =$
311 5–40, for the median over events of the homogeneous double-couple synthetics relative to the observed median.
312 Brune row: change of the homogeneous coda-to-S ratio relative to an impulsive source.

313 Table 4 shows that the deficit does not depend on the analysis choices. Ending the fit window
314 at $S + 10$ or $S + 20$ ms, or moving the coda window, changes the smooth-model medians by a
315 few tenths of a log unit. For the smooth media the window overlap of the original analysis had
316 little effect: with the same coda window, fitting to $S + 60$ instead of $S + 15$ ms changes the
317 medians by at most 0.02. The original coda window, which began at $S + 30$ ms, gave a
318 smaller deficit than the corrected one simply because it included more of the early coda.

319 Deviatoric and double-couple solutions give the same result, although they fit the direct
320 waves less well (median VR in the homogeneous medium 0.75 for the full, 0.71 for the
321 deviatoric and 0.67 for the double-couple solution). The moment-tensor problem is
322 moderately conditioned (median condition number 16–18 for smooth and 4–16 for stochastic
323 media, decreasing with σ). The full solutions carry large non-double-couple parts (median
324 $|\text{ISO}|$ 28 %, $|\text{CLVD}|$ 28 % in the homogeneous medium). With two sensors in one well these
325 parts are poorly resolved and we do not interpret them. The point is that no source constraint
326 changes the coda deficit. The observed S peak lies within ± 10 ms of the predicted time for
327 250 events on both sensors, and the deficit remains in that subset and in the ± 5 ms subset.

328 Timing and location errors therefore do not explain it.

329 The ringing tests constrain the instrument explanation (Fig. 5). Between 30 and 120 Hz the S
330 spectra of both sensors are smooth, and the coda/S spectral ratio has no peak. The worst
331 resonance compatible with this smoothness leaves the homogeneous medium 1.6–2.0 log

units short at sensor A and 1.0–1.3 at B. The resonances above 300 Hz are strong and clearly visible, so an in-band resonance large enough to create the observed coda would not have escaped detection. This test excludes a single resonant mode, but not a broadband, non-resonant coupling response with a long impulse response. That possibility can be excluded only with an empirical transfer function, which is not available.

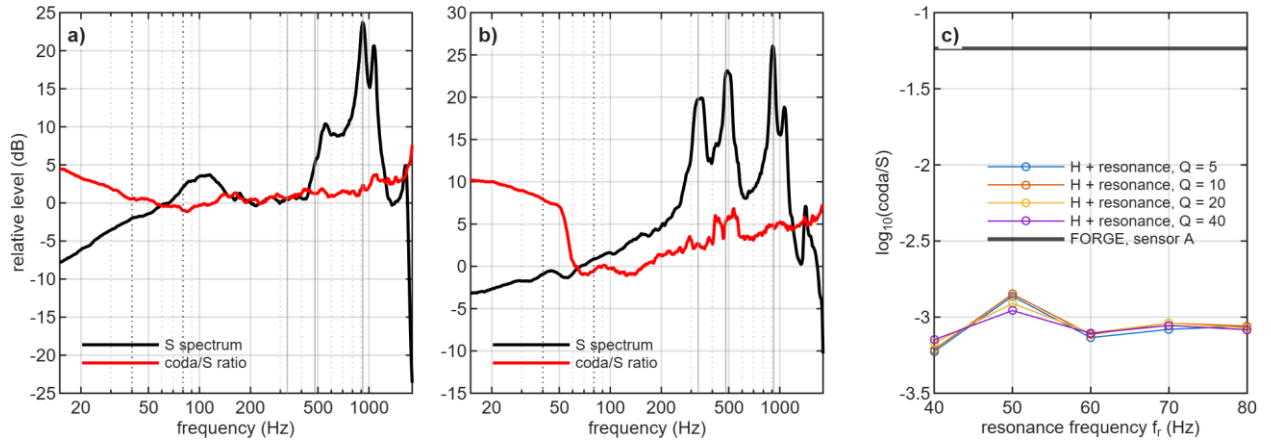


Fig. 5 Tests for instrument or borehole ringing

a, b) Sensors A and B: median S-window spectrum normalised over 30–150 Hz (black) and median coda/S spectral ratio (red, normalised). Dotted lines mark the analysis band 40–80 Hz and grey lines the resonances near 330, 480 and 920 Hz. c) Null-ringing test: log₁₀ coda/S of the homogeneous synthetics (sensor A) after convolution with the strongest resonance compatible with the observed in-band spectral smoothness, versus resonance frequency, for $Q = 5$ –40. Thick line: observed FORGE median.

Grid refinement and sponge thickness change the late energy little (Additional file 1: Table S10). Relative to the 5-m reference, the median coda-to-S ratio at 40–80 Hz changes by +0.01 log units on the 2.5-m grid and by +0.13 with the doubled sponge (+0.16 and –0.13 at 20–40 Hz). Enlarging the domain by 150 m on every side, however, raises it by +0.40 at 40–80 Hz (+0.09 at 20–40 Hz) and lowers the coda decay rate from 48 to 21 s^{–1}. Because the doubled sponge has a much smaller effect, this is not a boundary artefact. It reflects the larger scattering volume: in the reference domain the event cloud nearly fills the interior, so

heterogeneity beyond it lies in the absorbing sponge and returns no scattered energy. A smooth medium has no such scatterers, so the smooth-model deficit is unaffected. The slower decay in the larger domain may also explain why the observed coda decayed more slowly than the simulated coda in the original analysis. For the stochastic media, the reference domain probably underestimates the late energy, and the effective scattering strength derived below is probably an upper bound. With the sensitivity of the median ratio to σ found below (0.80 log units per unit change of $\ln \sigma$), this single test implies that σ_{eff} from the reference domain may be overestimated by a factor of about 1.7. Whether the same factor applies to other families and to sensor A was not tested.

Effective scattering strength

For each family the σ at which the median predicted late energy equals the observed one (σ_{eff}) is interpolated in $\ln \sigma$ between the simulated levels of the reference domain (Fig. 6; Additional file 1: Table S11). At sensors A and B it is 0.104 and 0.053 for pink media with $p = 1.5$, 0.098 and 0.048 with $p = 1.8$, 0.093 and < 0.09 for exponential media with $a = 15$ m, 0.102 and < 0.09 with $a = 50$ m, and 0.112 and 0.131 for Gaussian media with $a = 50$ m (" < 0.09 ": the smallest simulated σ already over-predicts). σ_{eff} thus differs between sensors and families, so it is a model-dependent summary of the late energy rather than a unique rock property. Media with less small-scale content need larger σ (Gaussian $a = 50$ m), and media richest in structure near the S wavelength need less (Gaussian $a = 15$ m: < 0.09 at both sensors). Realisation scatter adds to this. The two seeds at one σ differ in the median ratio by up to 0.47 log units for pink media and 0.40 for exponential media with $a = 15$ m, i.e. a factor of about 1.8 in σ_{eff} , but by up to 1.57 for Gaussian media with $a = 50$ m, whose few large features make each realisation idiosyncratic in a 0.5-km box.

Relaxing the constant V_P/V_S ratio changes the late energy only moderately. With independent V_P and V_S fields the median ratio is $-0.13/+0.54$ (A/B), against $+0.31/+0.78$ for the coupled

medium. The V_S -dominated medium, with half the V_P perturbation, gives $+0.32/+0.79$. The S coda is controlled mainly by the V_S perturbation, so σ_{eff} should be read as a V_S scattering strength.

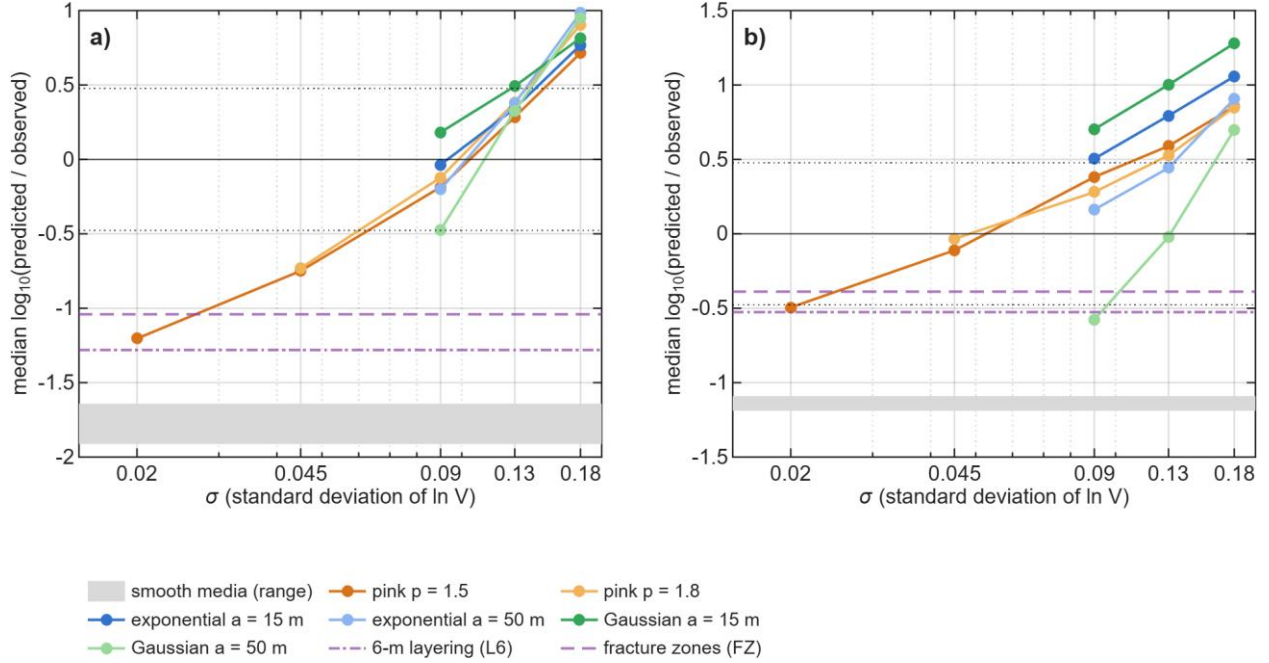


Fig. 6 Median predicted/observed late S energy versus σ for each spectral family

a) Sensor A. b) Sensor B. Median over events of $\log_{10}(\text{predicted}/\text{observed})$, averaged over realisations where several exist. The zero crossing defines σ_{eff} ; dotted lines: factor of three; grey band: range of the smooth media; purple dash-dotted and dashed lines: 6-m layering (L6) and fracture zones (FZ).

Inter-event coda coherence

Fig. 7 shows D2 in the three windows. In the early coda (S + 15 to S + 100 ms) the observed coherence falls from 0.69 at 0–5 m to 0.49 at 80–160 m (sensor A; 0.75 to 0.48 at B). The smooth media stay coherent (all pairs within 160 m: 0.82–0.86 at A, 0.73–0.78 at B, against 0.50 at both sensors for the data), and so do the two deterministic small-scale media (0.80–0.83 and 0.67–0.69). The stochastic media decorrelate as the data do (0.48–0.59 and 0.54–0.62). This window, however, contains the tail of the S pulse and is close to the window of the source fit.

394 In the late windows the picture changes. All coherences are lower and approach the coherence
 395 of noise alone (0.57 at A and 0.48 at B). The observed late coda is at or below this level (0.53
 396 and 0.45 for $S + 40$ to $S + 100$ ms; 0.54 and 0.45 for $S + 50$ to $S + 110$ ms). Because the
 397 synthetics carry the same noise, D2 in these windows measures how much coherent late
 398 energy a medium adds. The smooth media (0.60–0.69 at A, 0.53–0.58 at B) and the fracture
 399 zones (0.64, 0.62) remain more coherent than the data in both late windows; their 95 %
 400 intervals lie above the data interval in 19 of 20 medium–sensor–window cases (Additional file
 401 1: Table S17). The stochastic media (0.49–0.55, 0.49–0.50) and the 6-m layering (0.57, 0.52)
 402 come closest; in $S + 50$ to $S + 110$ ms the 6-m layering is as close to the data as the stochastic
 403 media at sensor A (0.51 against 0.50–0.52 and 0.54 observed). In the pure late coda D2
 404 therefore separates the data from the smooth media and from the planar fracture zones, but not
 405 from fine 1-D layering. The separation of distributed 3-D heterogeneity from fine layering
 406 rests on the early-coda window and is correspondingly less secure.
 407 Location error does not change these conclusions. Perturbing the hypocentres of data and
 408 media by up to 50 m flattens the binned curves, as expected, but changes the observed
 409 medians at separations of 20 m or more by at most 0.05 (Additional file 1: Fig. S10), and the
 410 all-pairs levels, which do not depend on the binning, are unaffected. We therefore do not use
 411 the 0–5 and 5–10 m bins for inference (the 0–5 m bin contains only 64 pairs from 96 events).
 412 D2 is measured on the same records as the late energy, so it is a second descriptor of the same
 413 data, not an independent experiment.

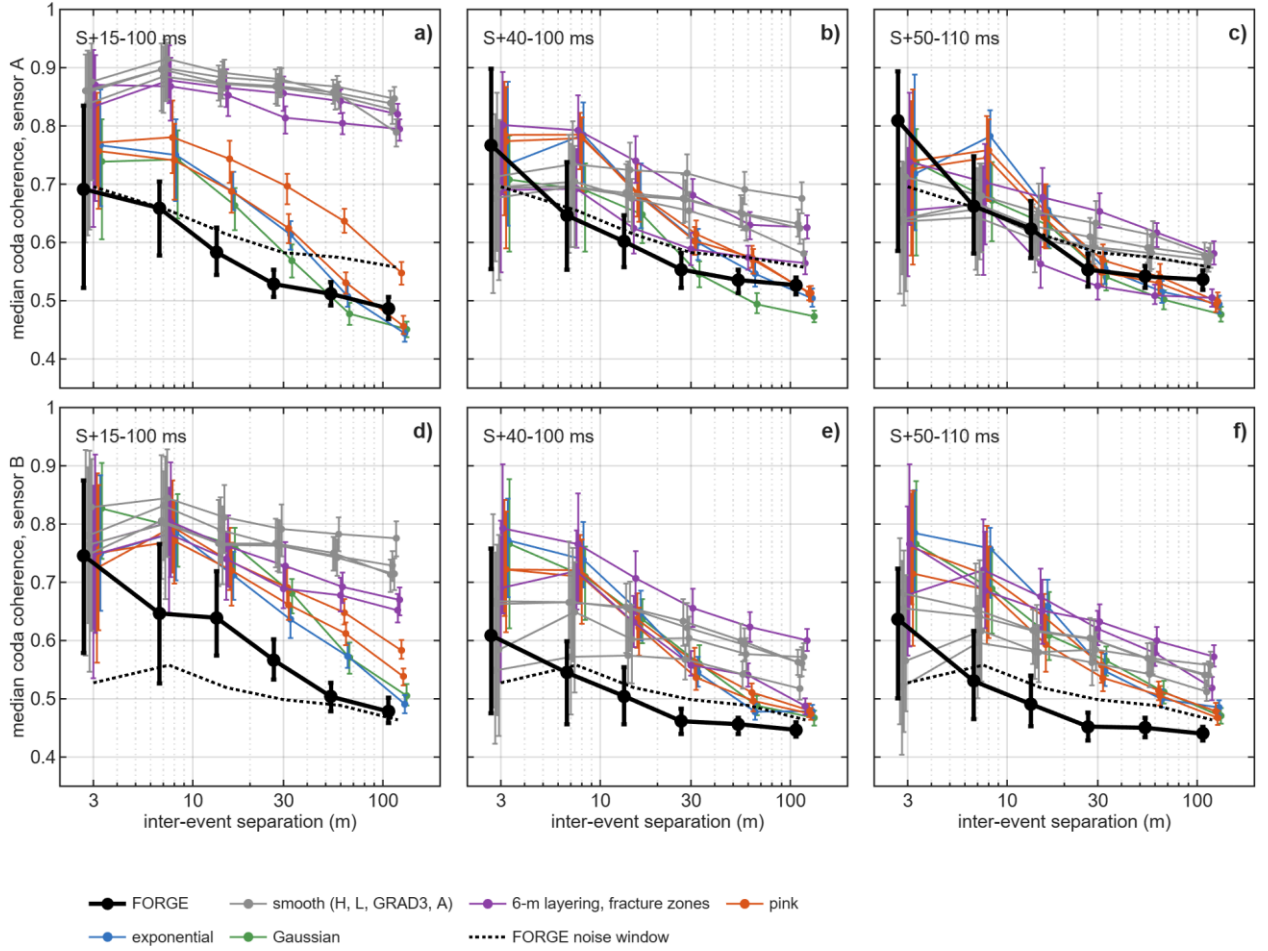


Fig. 7 Inter-event coda coherence in early and late coda windows

Median coherence of the 40–80 Hz coda versus inter-event separation, with 95 % cluster-bootstrap intervals over events, for sensor A (a–c) and sensor B (d–f), in the windows S + 15 to S + 100 ms (a, d), S + 40 to S + 100 ms (b, e) and S + 50 to S + 110 ms (c, f). Black: FORGE; dotted: FORGE noise-only window (S + 550 to S + 610 ms); grey: smooth media (H, L, GRAD3, A); purple: 6-m layering and fracture zones; orange: pink ($\sigma = 0.09, 0.13$); blue: exponential ($a = 15$ m); green: Gaussian ($a = 50$ m), $\sigma = 0.13$. Symbols are offset horizontally for clarity.

Statistical comparison and the spectral family

In the synthetic tests the statistical comparison selected the generating class in both controlled examples. With the pink truth T1 the best medium was a pink medium with the true σ ($R1, \Phi = 0.09$), ahead of H, L and A ($\Phi = 0.29$ each). With the layered-VTI truth T2, L and A ranked

427 first (0.13, 0.16) and every pink medium was rejected. Two examples do not establish that the
 428 procedure is unbiased. The deterministic moment-tensor fit behaved differently. For the pink
 429 truth it preferred the smoothest media (median VR 0.93 for H against 0.88 for a pink medium
 430 with the correct statistics but another realisation), because a wrong realisation places scattered
 431 energy at wrong times. This applies to selecting a model class from a set of arbitrary
 432 stochastic realisations. It does not imply that full-waveform inversion, which updates the
 433 structure itself, cannot recover heterogeneity when the data coverage allows it.

434 The equal- σ family test selected the correct family for 5 of 9 truths. Under a chance model
 435 with $p = 1/3$ the probability of at least 5 successes is 0.145, so this is not significant. Pink and
 436 exponential truths were assigned to the correct member of that pair in 3 of 6 cases (chance
 437 level), whereas small-scale-rich (pink or exponential) and Gaussian truths were separated in 8
 438 of 9. With so few realisations, even the latter is only indicative.

439 On the FORGE data (Fig. 8), the smooth media (best $\Phi = 1.33$) and the deterministic small-
 440 scale media (1.01) are separated from every stochastic family (best members 0.25–0.57) by
 441 far more than the bootstrap intervals. Between the stochastic families, the best single member
 442 belongs to the exponential $a = 50$ m family ($\Phi = 0.25$ [0.24, 0.30]), which has the lowest Φ in
 443 98 % of event-bootstrap resamples, ahead of the pink family (0.29 [0.27, 0.34]). In the
 444 holdout test the best training member scores a median held-out Φ of 0.26 (exponential $a = 50$
 445 m) and 0.31 (pink). Both tests, however, resample events with the realisation fixed, and the
 446 holdout selected the same realisation in almost every split. They show that the ranking
 447 generalises across events, not across realisations. The realisation matters more: the two seeds
 448 at the same (σ , shape) differ by up to 0.33 in Φ (median 0.11), whereas the best members of
 449 the leading families differ by 0.05. Averaged over both seeds, the best grid point of each
 450 family gives exponential $a = 50$ m 0.35 ($\sigma = 0.13$); pink 0.37 ($\sigma = 0.13$, $p = 1.8$); Gaussian $a =$
 451 50 m 0.46 ($\sigma = 0.13$); exponential $a = 15$ m 0.48 ($\sigma = 0.13$); Gaussian $a = 15$ m 0.58 ($\sigma =$

0.09). With two realisations per grid point, the 20–80 Hz statistics therefore do not select a spectral family; Gaussian media with $a = 15$ m fit least well, but even they are separated from the smooth and deterministic media by a margin (0.43 in Φ) larger than any realisation effect.

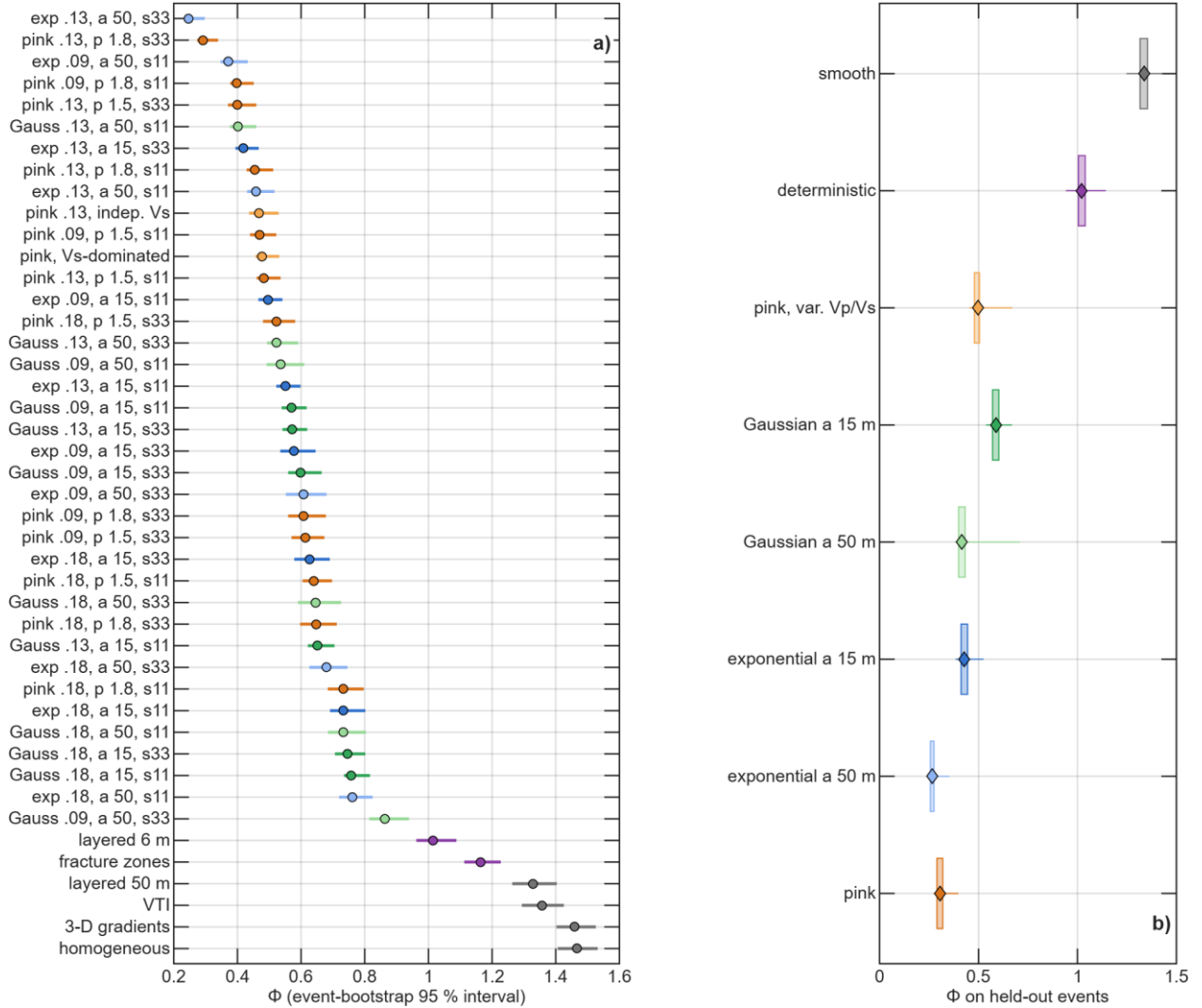


Fig. 8 Statistical misfit Φ with event-bootstrap intervals and holdout scores

a) Φ for all media with the full event set (circles) and 95 % event-bootstrap intervals (bars), sorted by Φ . Labels give family, σ , p or a (m) and white-noise seed (s11, s33). Grey: smooth media; purple: deterministic small-scale media; orange: pink; light orange: variable V_p/V_s ; blue: exponential (dark $a = 15$ m, light $a = 50$ m); green: Gaussian (dark $a = 15$ m, light $a = 50$ m). The intervals reflect event resampling only, not realisation scatter. b) Held-out Φ of

each family's best training member over 21 splits in both directions (box: interquartile range; whiskers: range).

Sonic-log spectrum

Fig. 9 shows the log spectrum with four fitted spectral models. The spectrum is consistent with a power law over 12–200 m, but this range spans only 1.2 decades. Short-correlation-length spectra fit almost as well, and the estimated slope depends on estimator and segment. The log therefore does not select a spectral family independently either.

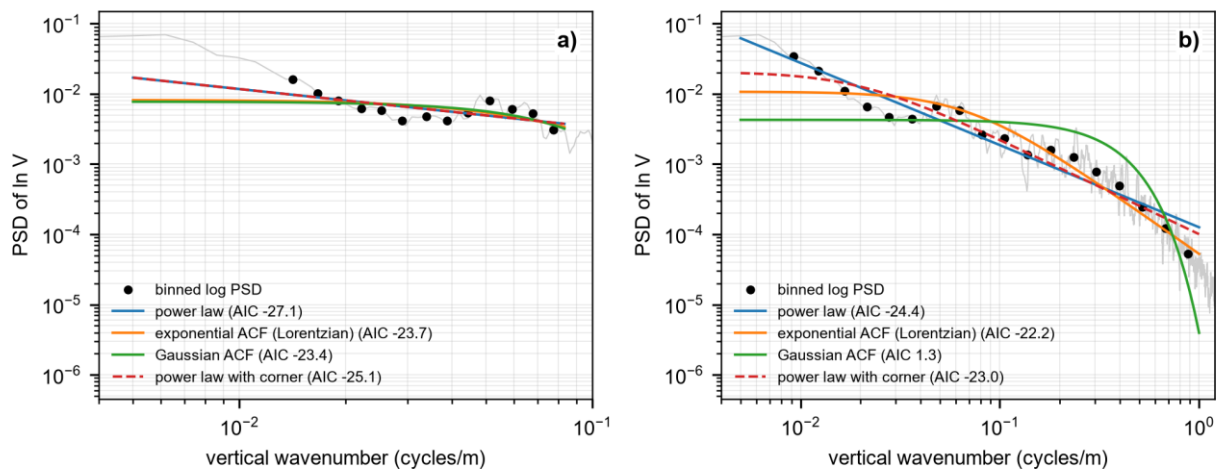


Fig. 9 Spectral models of the 56-32 sonic-log fluctuations

Multitaper power spectrum of detrended $\ln V$ (grey) and log-binned values (dots), with least-squares fits of a power law, exponential and Gaussian autocorrelation spectra and a power law with a corner (dashed). a) 12–200 m; the corner model converges to a pure power law and overlies it. b) 1–200 m. AIC values in the legend.

Discussion

What the data show

Media without structure below ~ 50 m (homogeneous, 50-m layering, smooth 3-D gradients, VTI) produce one to two orders of magnitude less late S energy than recorded, event by event. This deficit survives non-overlapping windows, alternative source constraints, finite source

duration, S-timing selection and a bounded single-resonance ringing model, and the simulated coda is converged with respect to grid and absorbing boundary. The two simple deterministic media with structure at the 5–10 m scale that we tested, the logged layering at full resolution and twelve thin planar fracture zones, produce much of the missing energy, particularly at the deeper sensor. In the early coda such structure keeps neighbouring events coherent, whereas the observed coda decorrelates as in the stochastic media. In the pure late coda the planar fracture zones and the smooth media remain too coherent, but the 6-m layering and the stochastic media both approach the noise-limited coherence of the data. What remains true if the term "pink noise" is removed from this paper is therefore: within elastic modelling, the late S energy in the 56-32 records requires structure at the 5–10 m scale, and its decorrelation favours distributed 3-D heterogeneity over smooth media and planar fracture zones, without excluding fine layering, unless an unmodelled site or coupling response of comparable size is present.

What the data do not show

First, the waveforms do not identify the spectral family. At 20–80 Hz, power-law, exponential and Gaussian media reach comparable agreement once σ is adjusted, and the equal- σ truth tests do not separate pink from exponential media. The original analysis had suggested that the 100-Hz coda level (2.5-m grid, sensor B) rejects a large-scale Gaussian medium. That comparison used fixed $\sigma = 0.13$, two Gaussian realisations and noise-free synthetics, and changed grid, domain and sensor at once. We therefore report it only as an exploratory result (Additional file 1: Fig. S9). The same applies to the coda-attenuation diagnostics (Q_c^{-1}): at 25 Hz the coda window spans one to three periods and even the homogeneous medium yields a non-zero Q_c^{-1} . We treat them as finite-window diagnostics and draw no inference from the number of octave bands (Additional file 1: Figs. S7, S8).

504 Second, σ_{eff} is not a unique measure of rock heterogeneity. It absorbs every unmodelled
 505 contribution to the late energy: density contrasts, anisotropic or fluid-filled fracture scattering,
 506 intrinsic attenuation (which would reduce the coda and raise σ_{eff}), and site or coupling effects
 507 that the ringing test cannot exclude. It also depends on the simulation domain. In the reference
 508 domain $\sigma_{\text{eff}} = 0.05\text{--}0.13$, depending on sensor and family. One domain test (one medium,
 509 sensor B) suggests that these values may be about 1.7 times too large. In the original analysis,
 510 the misfit optimum ($\sigma \approx 0.18$) and the per-event optimum ($\sigma \approx 0.09\text{--}0.13$) disagreed with the
 511 sonic log (0.045 at 6-m averaging) by up to a factor of four. With non-overlapping windows,
 512 all events, separate sensors and the domain effect, the discrepancy is much reduced. Adding
 513 the domain effect to the pink medium with the log value ($\sigma = 0.045$) gives median ratios of
 514 -0.33 (A) and $+0.42$ (B), within a factor of about 3 at both sensors. This agreement rests on
 515 one domain test at one sensor and should be read as consistency, not as a measurement.
 516 Third, the log and the wavefield sample different volumes: one vertical line through intact
 517 rock outside the stimulated zone, and the paths through the zone being stimulated. Stimulation
 518 might raise the scattering strength by opening fluid-filled fractures, which lower V_s
 519 preferentially. This is a hypothesis. A stage-by-stage comparison is not possible, because 402
 520 of the 416 events belong to stage 3 (the 14 stage-2 events show no significant difference;
 521 Additional file 1: Table S8). No pre-stimulation waveforms exist for these paths. The V_s -
 522 dominated medium shows only that a V_s -weighted perturbation would produce similar late
 523 energy, and the fracture-zone medium shows that twelve planar zones are not enough; neither
 524 shows that stimulated fractures produced the coda. Denser 3-D fracture networks, rough
 525 interfaces, irregular damage zones and anisotropic compliance were not tested, and a
 526 sufficiently dense deterministic network would approach a random medium.

Implications

For microseismic practice, the result means that velocity models adequate for locating events at ~ 1 -ms travel-time precision leave most of the late wavefield unexplained. Fine deterministic layering can supply much of its level, and its spatial incoherence is matched best, though not uniquely, by distributed 3-D heterogeneity. The late wavefield carries information on small-scale structure, but it must be compared statistically, because the realisation cannot be recovered from two sensors. For the GFI concept, the result is consistent but not decisive. Strong small-scale heterogeneity is required, and a lognormal power-law medium with roughly the logged σ provides it. The present seismic data do not, however, distinguish it from media with a short correlation length, nor fully from fine layering. Decisive tests of spectral shape need records at more stations, measured sensor transfer functions, larger simulation domains, and simulations above 100 Hz, where the families differ most. The 4-kHz FORGE records contain S energy above noise up to about 1.8 kHz (Additional file 1: Figs. S5, S6), but fixed-frequency tool resonances near 330, 480 and 920 Hz currently prevent interpreting that band as source or path energy.

Limitations

- Two sensors in one well, with unoriented horizontals and unreliable vertical channels; no calibration shots, perforation shots or independent sensors for 2022, so the sensor–borehole transfer function is not measured.
- Sensor positions, clock offsets and V_P trade off with the catalogue locations (~ 50 m, $\sim 8\%$).
- The stochastic media are isotropic, lognormal and elastic, with constant density; intrinsic attenuation, anisotropic fracture compliance and rough interfaces are not modelled.
- The domain-size effect on the stochastic coda was measured for one medium and one sensor; the correction of σ_{eff} is approximate.

- The common grid has two realisations per grid point, fewer than a full ensemble; realisation scatter limits σ_{eff} to about a factor of 1.8 for pink and short- a media and much more for large- a Gaussian media.
- The analysis band is 20–80 Hz, where the spectral families differ least.
- In the pure late coda the inter-event coherence is limited by noise and does not separate fine layering from distributed heterogeneity.

Conclusions

- Media without structure below ~ 50 m (homogeneous, 50-m layering, 3-D gradients, VTI) under-predict the late S energy of 416 FORGE MEQs by one to two orders of magnitude when the moment tensor is fitted to the direct waves only. The deficit is robust to windows, source constraints, source duration, event timing and a bounded in-band resonance, and the simulated coda is converged in grid and boundary.
- The tested simple deterministic 5–10 m structures (6-m sonic-log layering, twelve thin fracture zones) supply much of the late energy. In the early coda they keep neighbouring events more coherent than observed; in the pure late coda, where coherence is limited by noise, only the smooth media and the fracture zones remain distinguishable from the data.
- Stochastic 3-D media span the observed late energy and reproduce the loss of inter-event coherence more closely than the other tested media, with an effective scattering strength $\sigma_{\text{eff}} = 0.05\text{--}0.13$ in the reference domain. In one sensor-B sensitivity test, enlarging the domain implied that these values may be overestimated by a factor of about 1.7, which would bring them close to the sonic-log value of 0.045.
- Within 20–80 Hz the waveforms do not distinguish power-law, exponential and Gaussian spectra; the sonic log is compatible with a power law but does not exclude a short correlation length.

- Deterministic comparison against unknown stochastic realisations favours smooth media, so medium-class selection should use wavefield statistics.

Supplementary information

Additional file 1: Figures S1–S11 and Tables S1–S18 (reciprocity validation, sonic log, earlier statistical tests, bandwidth and multi-octave diagnostics, location stress test of the coda coherence, window and moment-tensor sensitivity, S-timing subsets, calibration bootstrap, source spectra, ringing tests, stage comparison, field statistics, numerical convergence, σ_{eff} curves, complete Φ and holdout tables, equal- σ truth tests, log-spectrum variants, per-event results for every medium and seed, and coda-coherence tables).

Availability of data and materials

The GES 2022 event catalogue is publicly available [10]. The 2022 56-32 SEG-2 downhole records are available from Utah FORGE and Geo-Energie Suisse on reasonable request. All code (FD solver, reciprocity, feature extraction, tests and figure scripts, with fixed random seeds) and all derived result tables are available at https://github.com/maswiet/Paper2_PinkNoise_WaveformInversion, release v1.0-submission (commit 1dab05b).

Competing interests

The authors declare that they have no competing interests.

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597 **Authors' contributions**

598 WS: conceptualisation, methodology, software, formal analysis, investigation, data curation,
599 visualisation, writing – original draft. PL: conceptualisation of the geoflow-imaging (GFI)
600 framework and of the pink-noise crustal model, methodology, writing – review and editing.
601 Both authors read and approved the final manuscript.

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